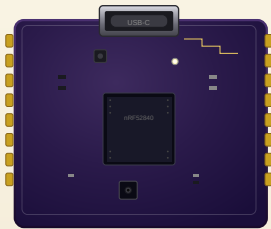


1

LAY OUT YOUR PARTS

These six things go on your desk before anything else.



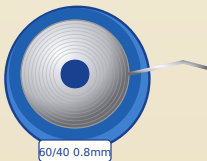
XIAO BOARD
(use 1, have 9 spares)



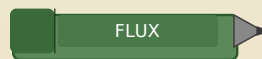
BATTERY
(red + black wires)



SOLDERING IRON
(don't plug in yet)



SOLDER WIRE
(60/40, 0.8mm)



FLUX PEN
(green marker thing)

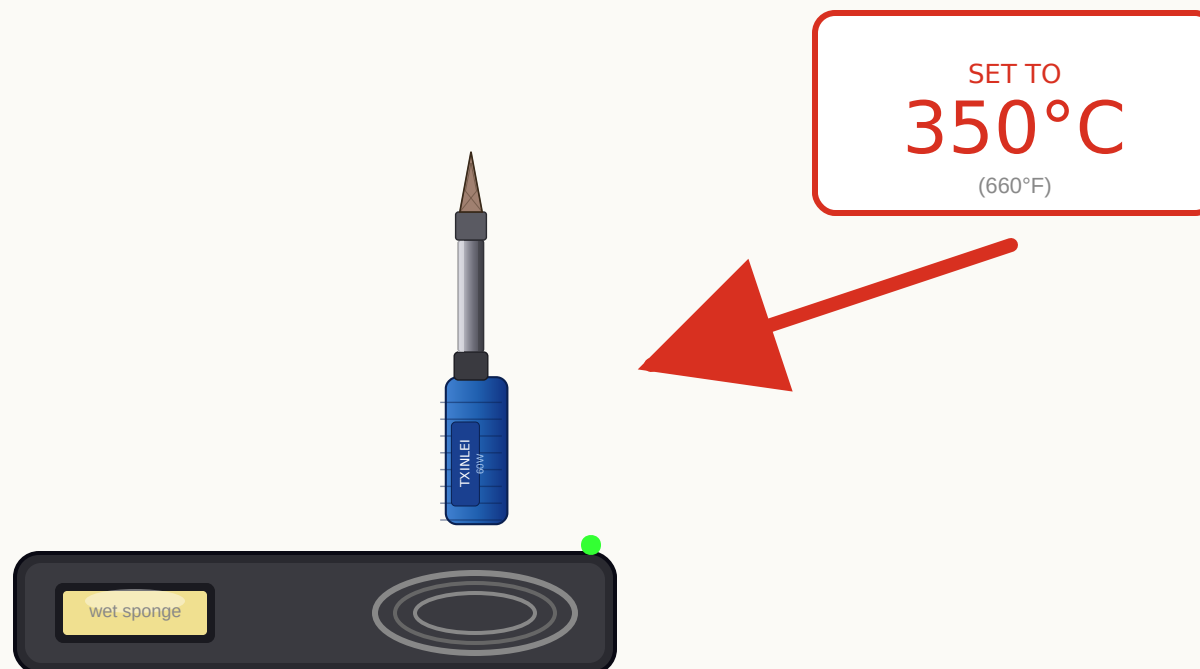


ORANGE TAPE
(Kapton, heat-proof)

2

PLUG IN THE IRON

Put it in the stand. Set the dial to 350°C. Wait 5 minutes.

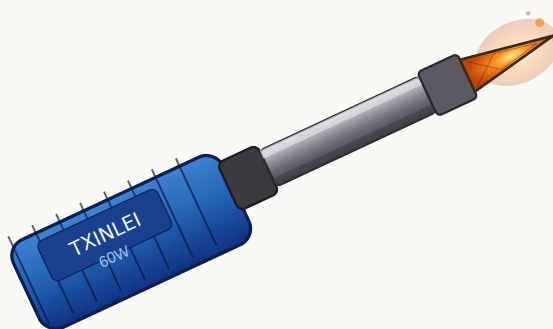


⏱ WAIT 5 MIN

3

⚠ SAFETY

The iron tip is hot enough to burn through skin in 1 second.



NEVER TOUCH THE METAL PART

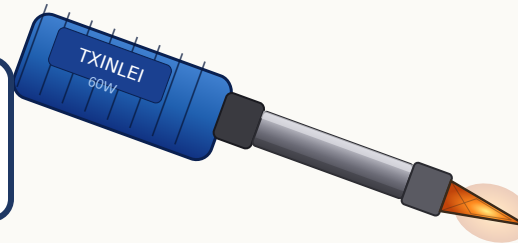
4

WIPE TIP ON WET SPONGE

Just brush it across. It'll sizzle. That's right.

WIPE THE TIP

Should sizzle and look
shiny silver after.



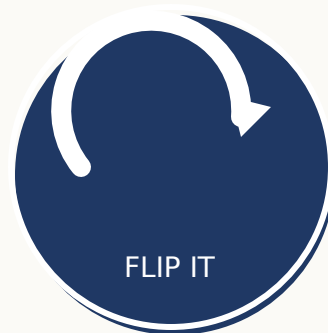
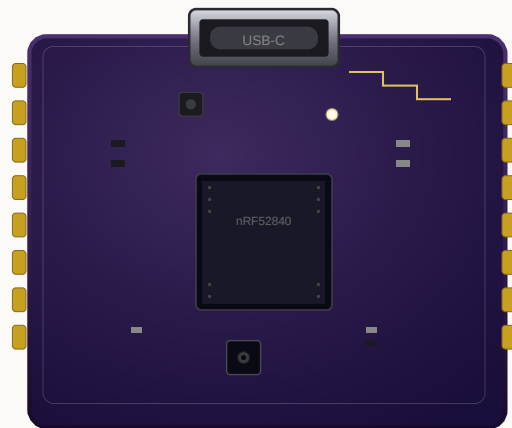
wet sponge

5

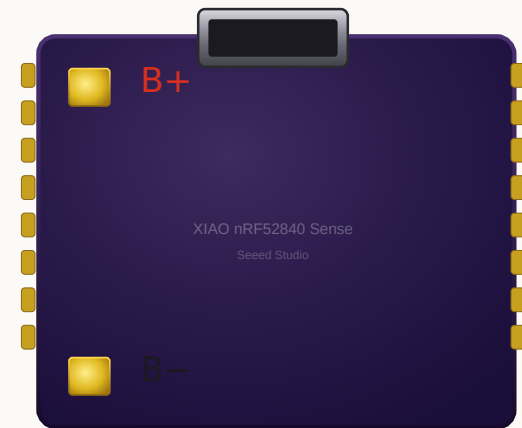
FLIP THE BOARD OVER

We need to see the BACK. That's where the battery pads are.

FRONT



BACK

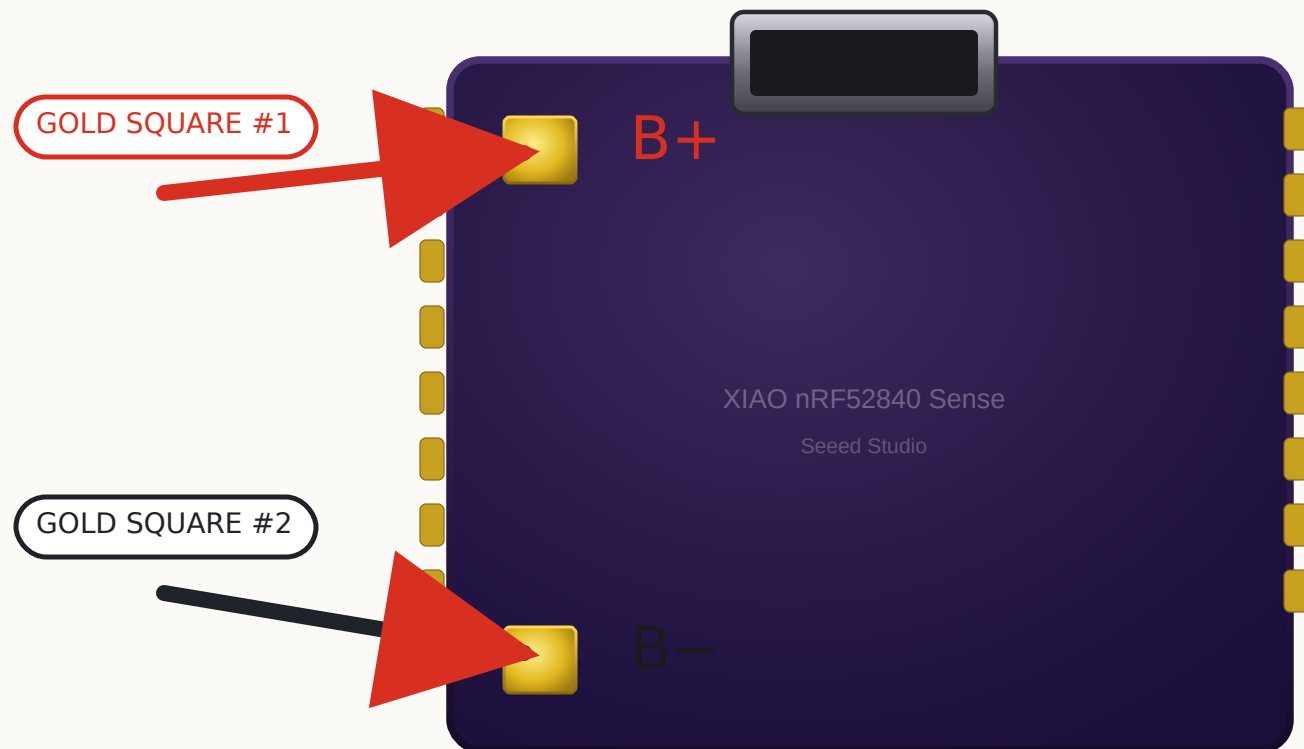


USB-C END POINTS UP, BOTH SIDES

6

FIND THE TWO GOLD SQUARES

These are the battery pads. Where the wires get attached.

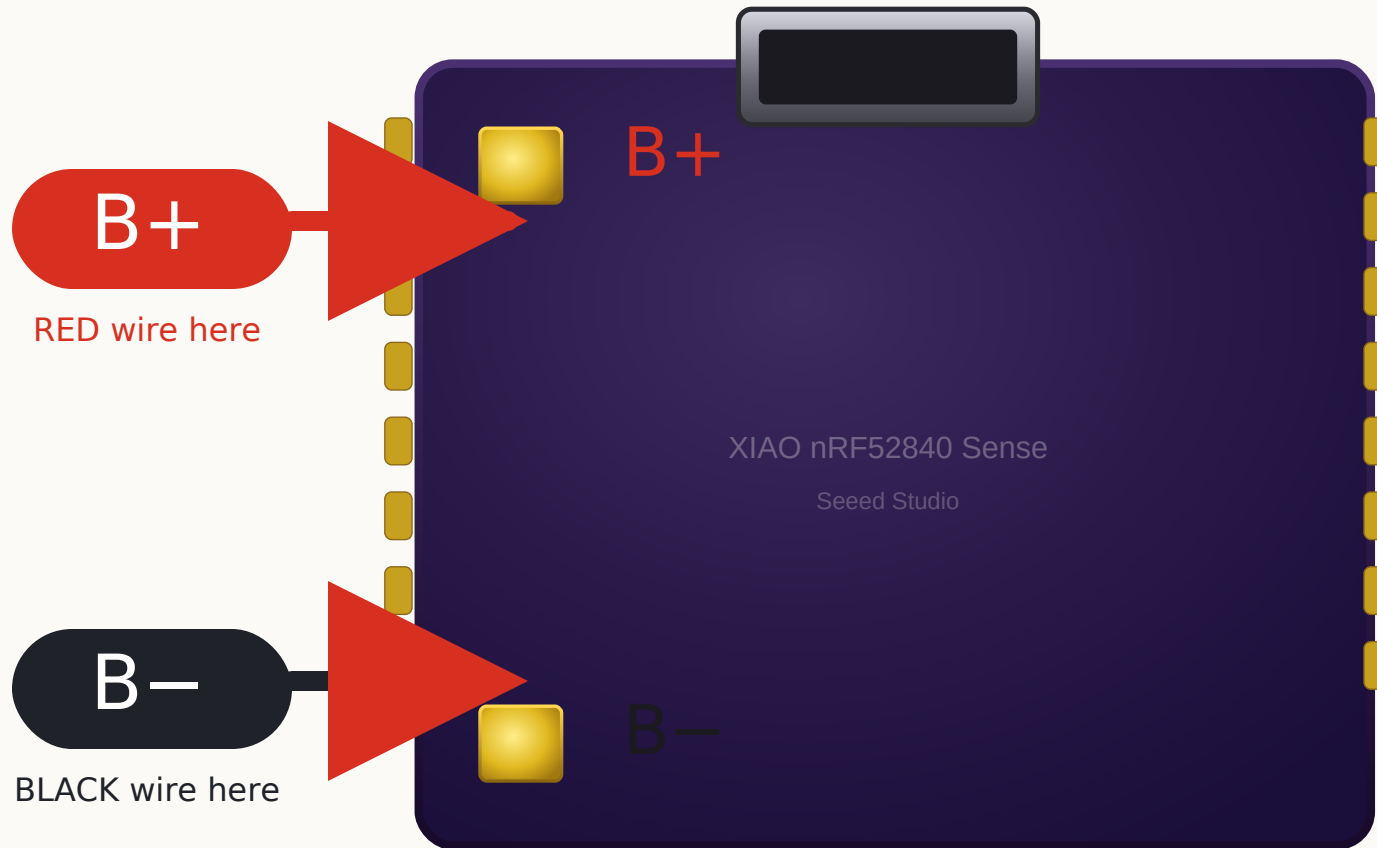


TWO TINY GOLD SQUARES. NEAR THE USB-C END.

7

B+ AND B-

B+ is for the RED wire. B- is for the BLACK wire.

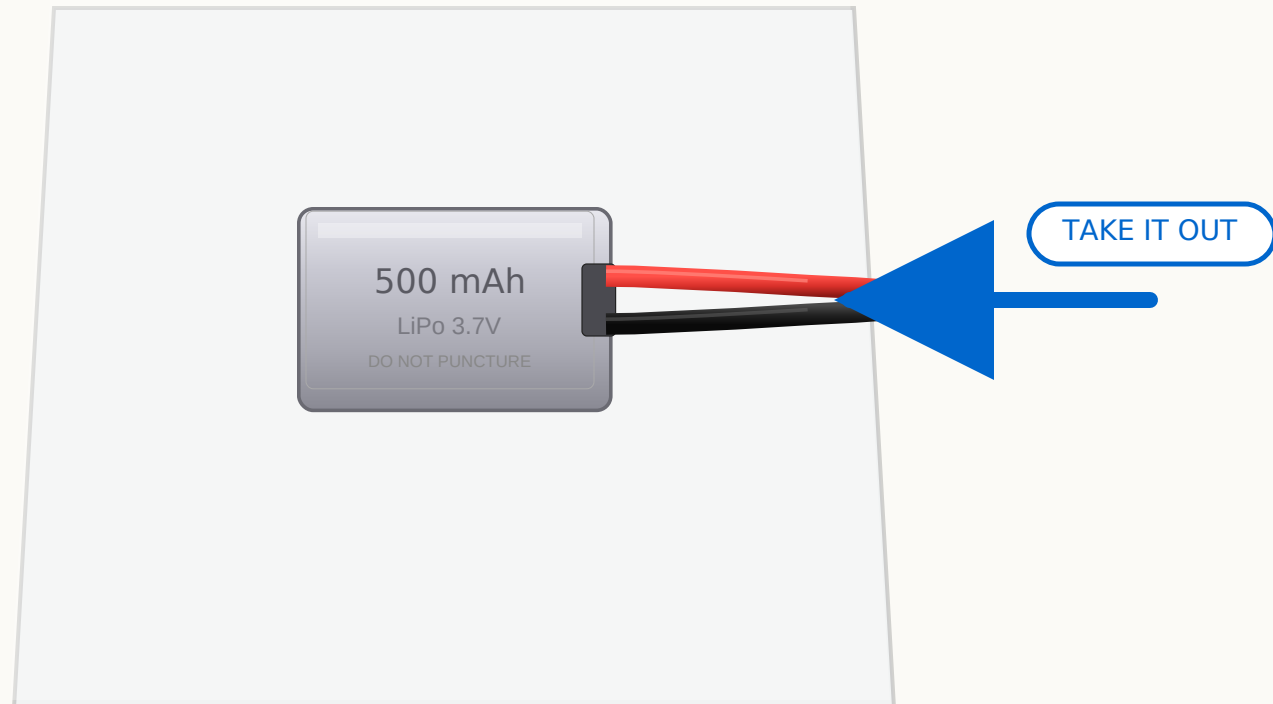


⚠ DO NOT MIX UP. RED → B+. BLACK → B-. Always.

8

GET THE BATTERY OUT

Open the baggie. Take one battery out. Set the rest aside.

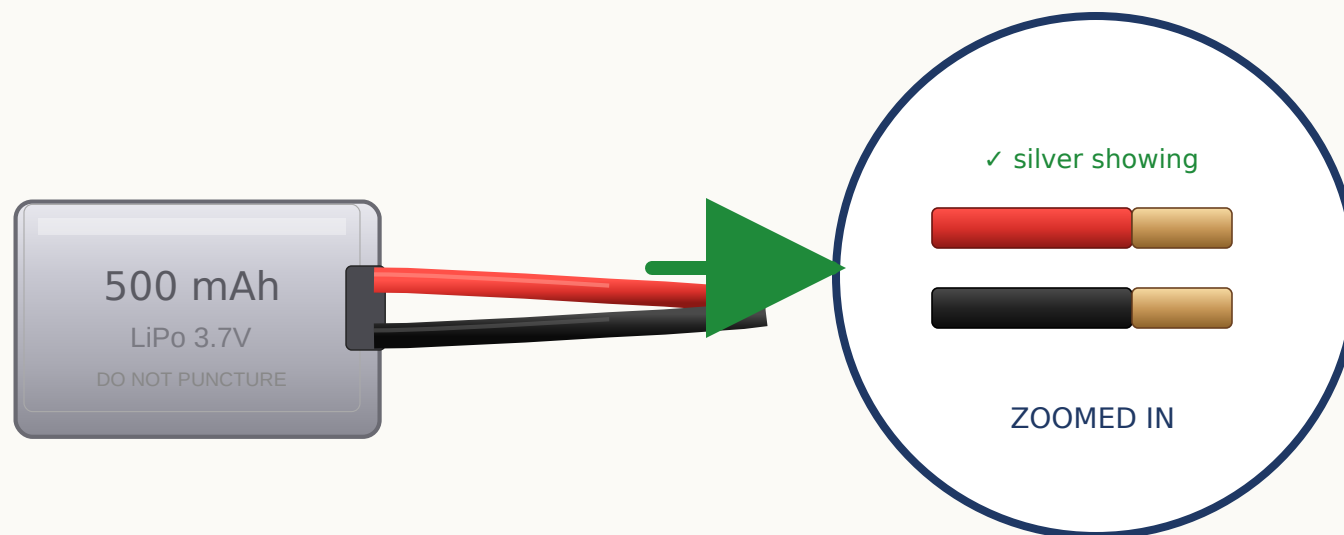


PICK ONE BATTERY. THE 500mAh DRONE LiPo.

9

CHECK THE WIRES

Look at the wire tips. About 3mm of bare silver should be showing.



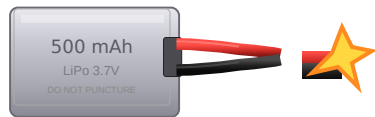
✓ THE BARE SILVER/COPPER TIPS SHOULD ALREADY BE SHOWING

10

NEVER LET WIRES TOUCH

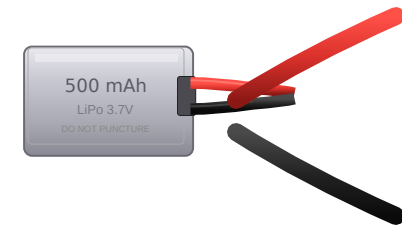
Bare red touching bare black = short circuit. Always keep them apart.

BAD ✗



wires touching = battery gets hot

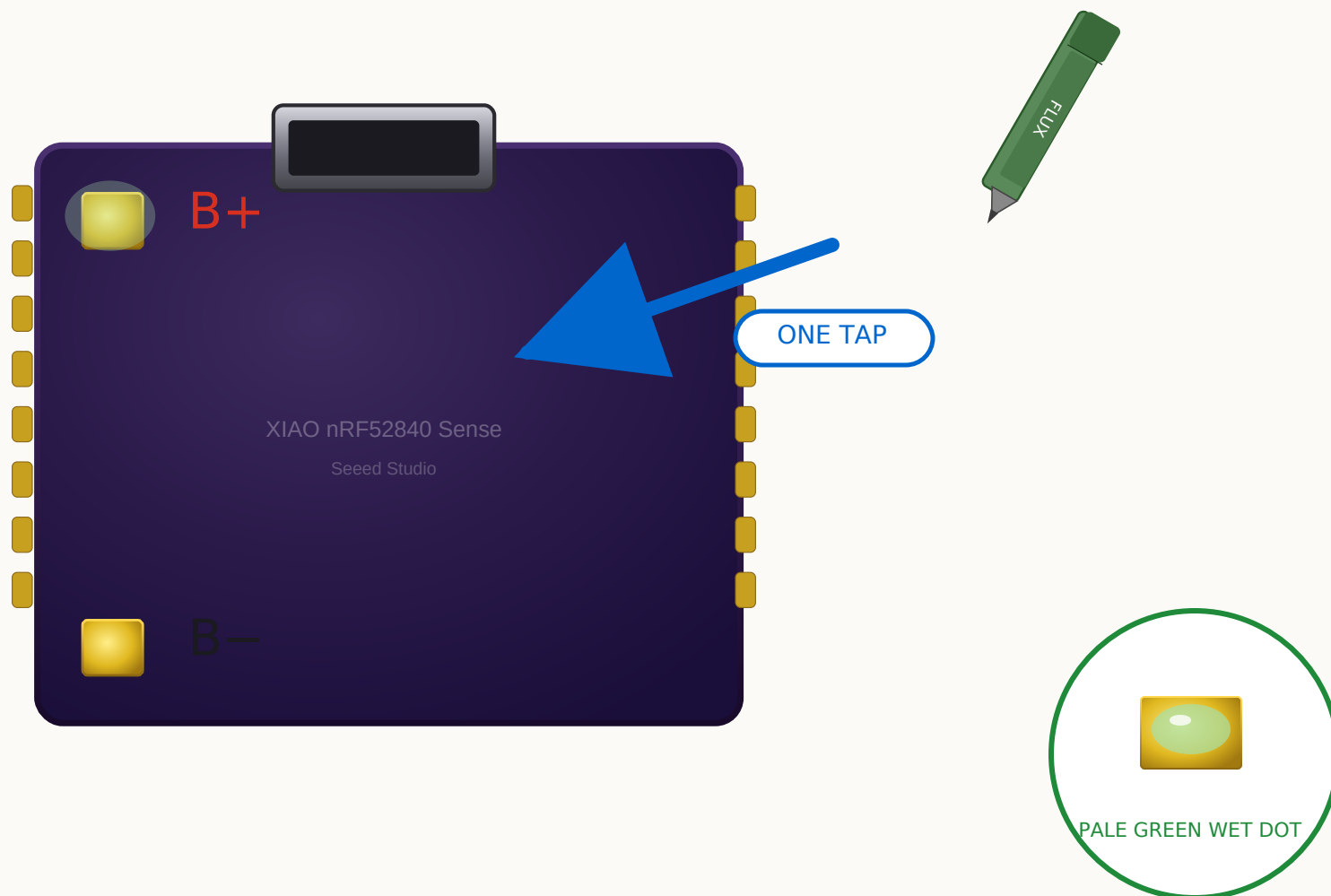
GOOD ✓



wires apart = safe

DAB A DOT OF FLUX ON B+

One tap with the flux pen. It looks like a tiny green puddle.



12

PLACE RED WIRE ON B+

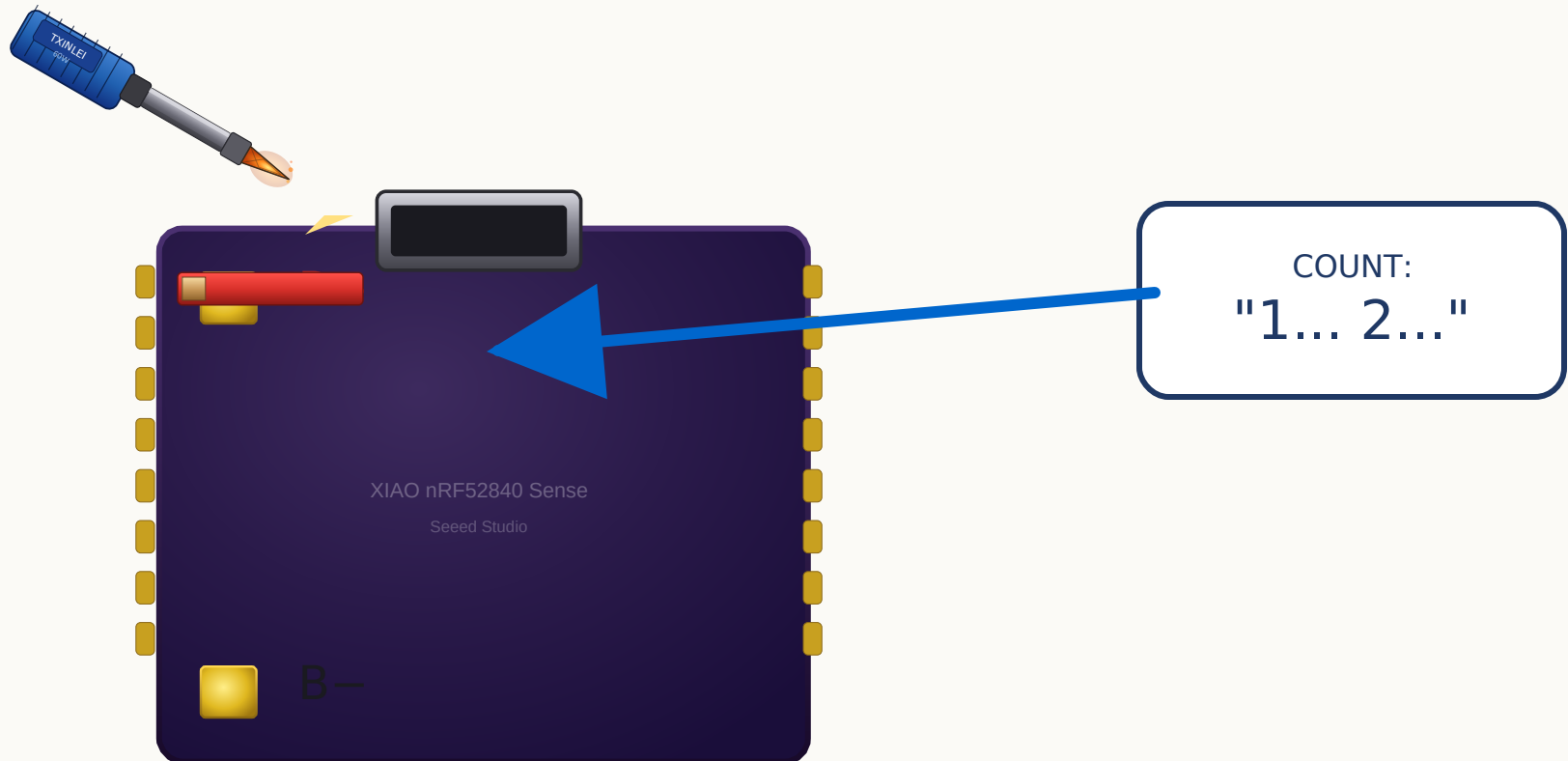
Lay it down. Bare silver tip touches the gold square. Hold it in place.



BARE COPPER TIP MUST TOUCH THE GOLD PAD

TOUCH IRON TO WIRE+PAD

Tip presses where the wire meets the gold pad. Count one, two.



14

FEED SOLDER INTO THE JOINT

Touch solder wire to the iron tip. A shiny blob forms. Pull both away.

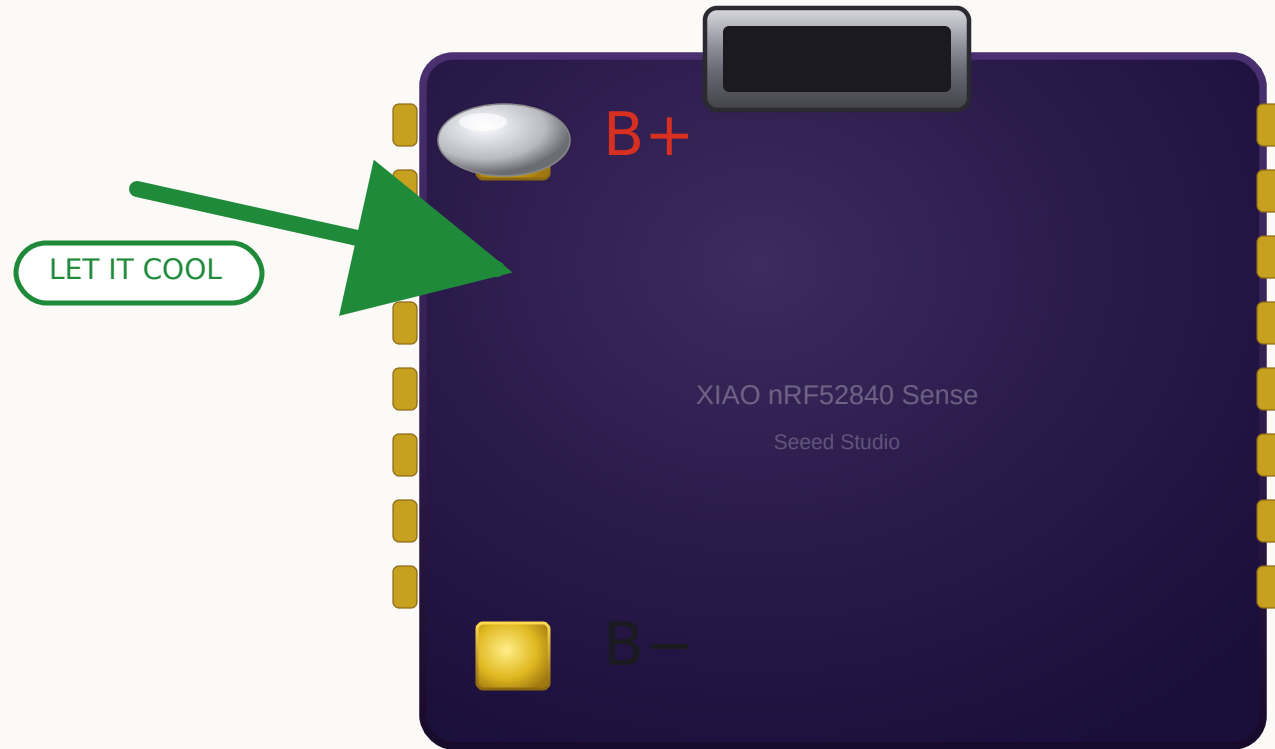


TOUCH SOLDER TO IRON. IT MELTS. PULL BOTH AWAY.

15

DON'T MOVE FOR 5 SECONDS

The solder is liquid for a moment. Then it locks solid.



□ HOLD STILL 5 SEC

WHAT GOOD LOOKS LIKE

Shiny silver dome. Smooth. Wider at the bottom than the top.



GOOD ✓

SHINY DOME, SMOOTH

✓ MIRROR SHINY = ELECTRICITY FLOWS

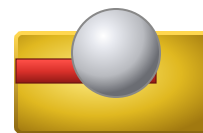
WHAT BAD LOOKS LIKE

Dull gray means it didn't bond. A ball means too much solder.



DULL / GRAY

→ reheat 2 sec



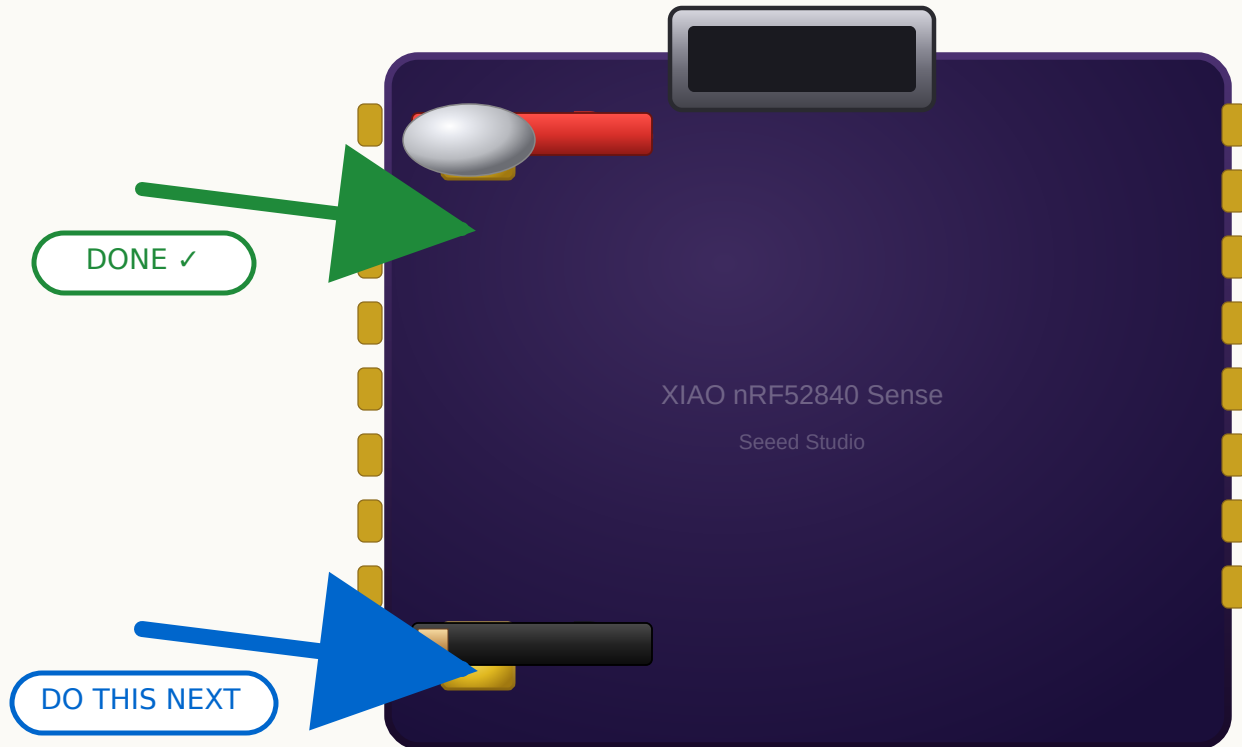
BALL ON TOP

→ flux + reheat

IF EITHER OF THESE → REHEAT AND TRY AGAIN

NOW SOLDER THE BLACK WIRE

Same process. Different pad. Different wire. B- gets the black one.



SAME EXACT STEPS: FLUX → LAY → TOUCH → SOLDER → COOL

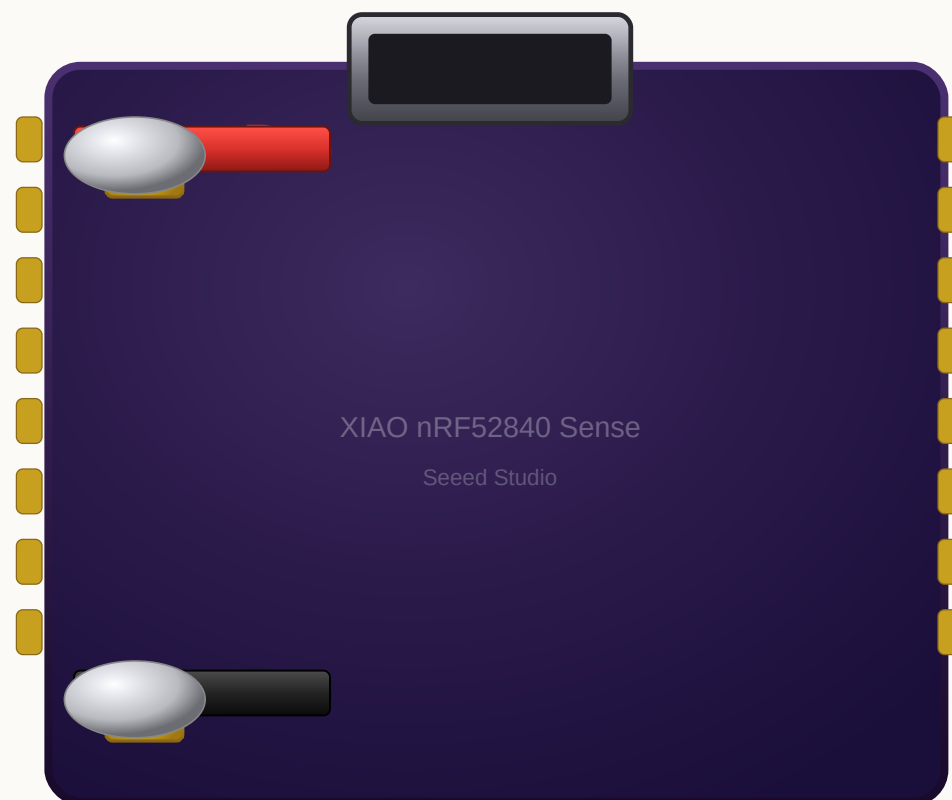
19

BOTH WIRES SOLDERED

Two shiny silver domes. One on B+ red. One on B- black.



BOTH
SOLDERED

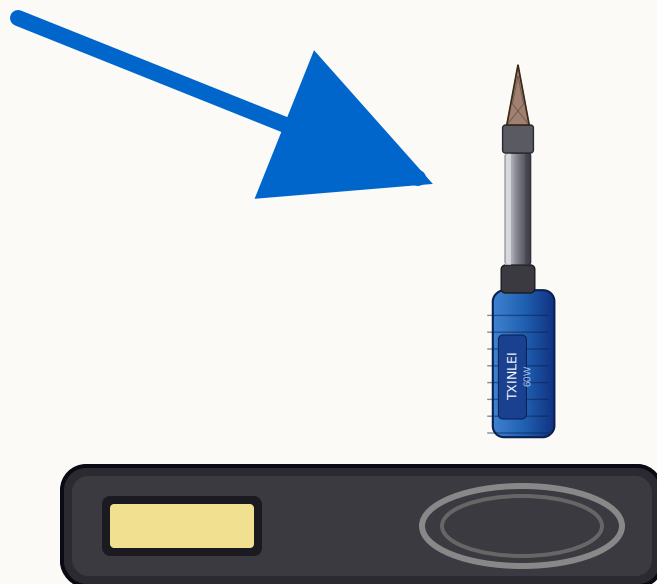


☑ ELECTRONICS DONE

PUT THE IRON BACK

Always in the stand. Even when you're 'just resting' it. Always.

BACK IN THE STAND

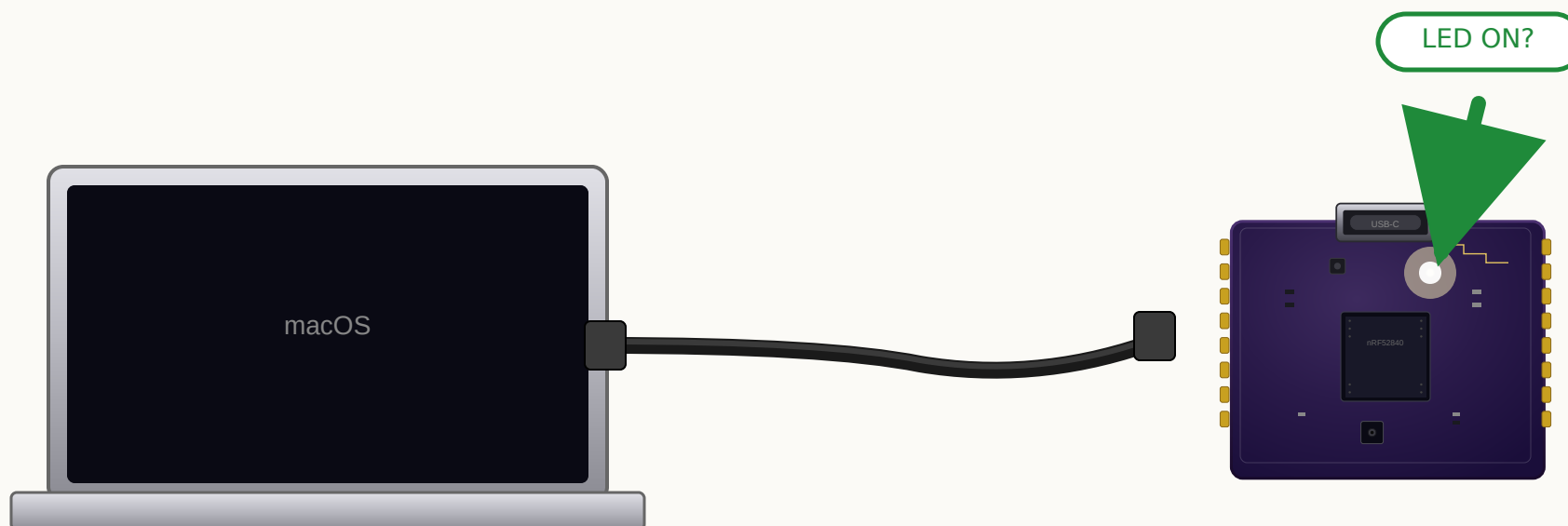


⚠ NEVER LAY THE IRON ON THE DESK. EVEN COOL, IT BURNS.

21

PLUG INTO YOUR MAC

USB-C cable. Mac. Look for a tiny light on the board.

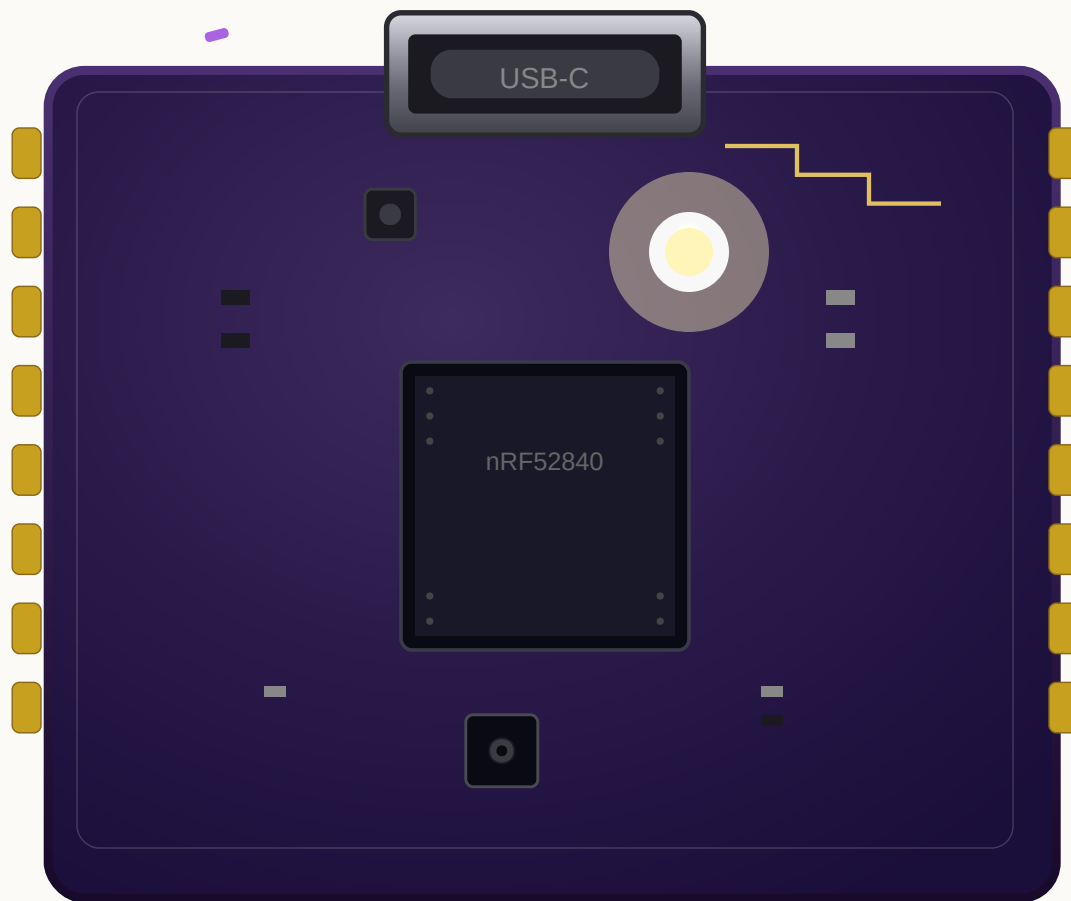


LIGHT ON = YOU DID IT CORRECTLY ✓

22

LED GLOWS = CELEBRATE

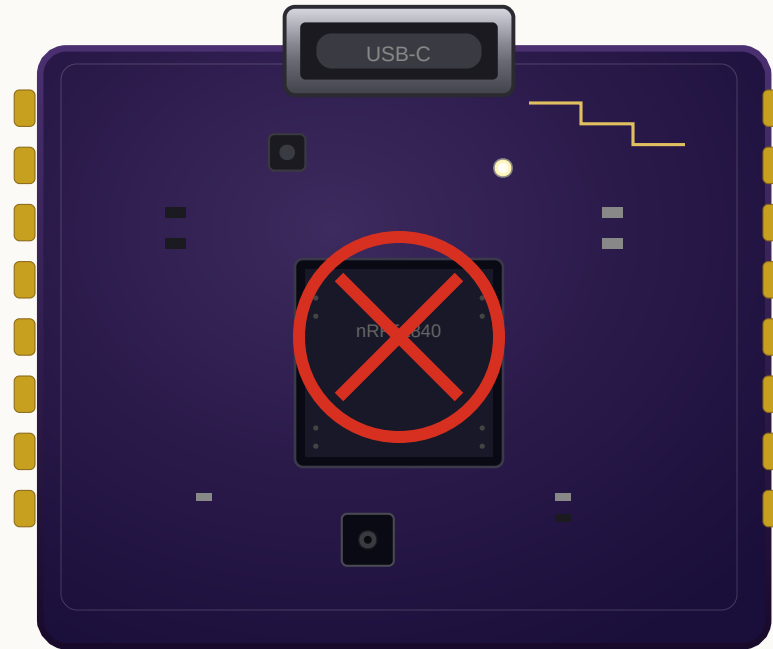
You built electronics. Unplug it now. Time to put a body on it.



☐ IT WORKS

IF NO LIGHT

Don't panic. Unplug. Check the three things below.



1. UNPLUG IMMEDIATELY.
2. CHECK: red wire on B+? black on B-? Not swapped?
3. CHECK: red and black wires NOT touching each other?

STILL NOTHING? USE A FRESH XIAO. YOU HAVE 10.

24

CUT A SMALL PIECE OF FOAM TAPE

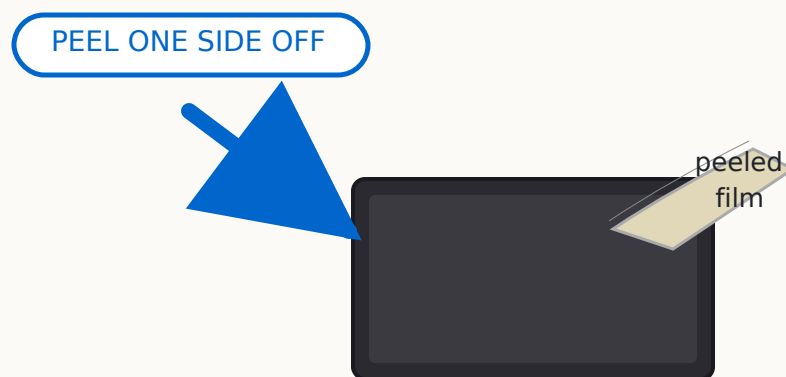
Black double-sided tape. Cut a piece about the size of a pinkie nail.



DOUBLE-SIDED FOAM TAPE FROM YOUR PARTS KIT

PEEL ONE SIDE OF THE TAPE

Pull off the paper film. Sticky side is now exposed.



↑ STICKY SIDE NOW EXPOSED ↑

26

STICK TAPE TO BATTERY

Press the exposed sticky side onto the flat side of the battery.

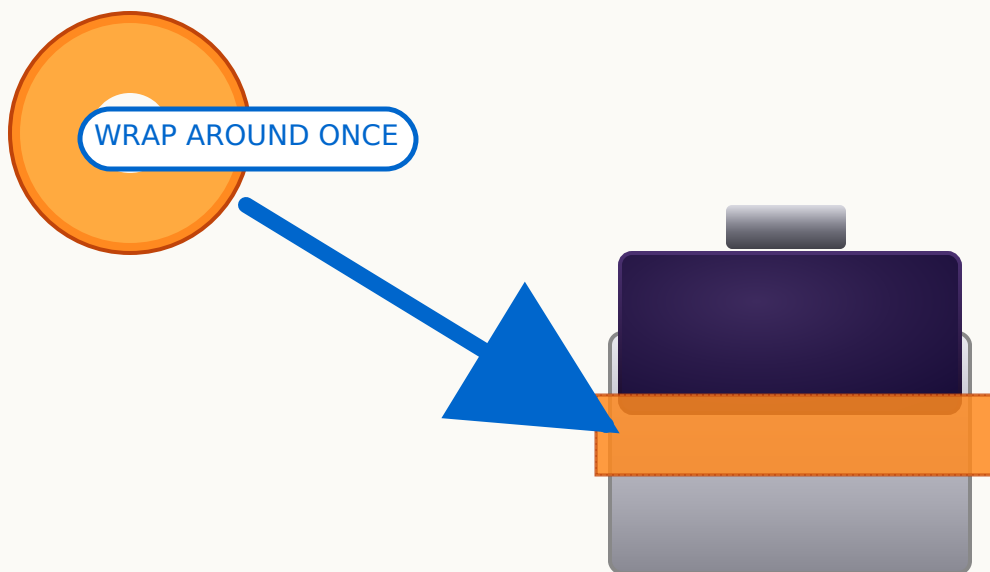


STICK IT TO THE FLAT SIDE OF THE BATTERY (NOT THE WIRE END)

28

WRAP IT IN ORANGE TAPE

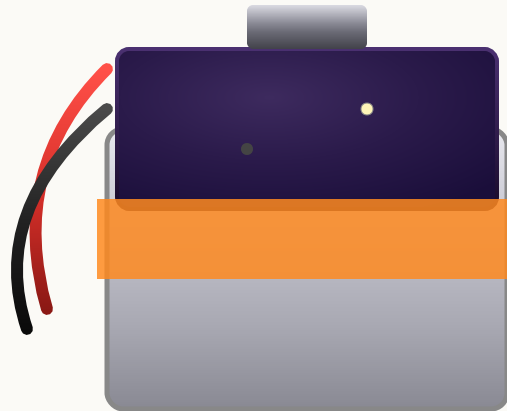
One lap of Kapton tape around the whole stack. Keeps everything tight.



ONE LAP OF ORANGE TAPE LOCKS IT ALL TOGETHER

DONE: ELECTRONICS COMPLETE

Charge it once now. Then we put a body on it.



YOU HAVE A
WORKING BRAIN

about 25mm × 22mm × 8mm
smaller than a quarter

⚠ TREAT IT LIKE A TINY PHONE. NO DROPS. NO WATER.